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Pre-registration: Does Japanese phonology count nasals beyond two?

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Introduction and theoretical background

One issue that is actively discussed in contemporary phonological studies is whether or not phonological systems can count only up to two. The predominant view in the generative literature has been that linguistic systems, including phonological systems, may count up to two but not more (e.g., Goldsmith, 1976; Hewitt & Prince, 1989; Ito & Mester, 2003; McCarthy & Prince, 1986; Myers, 1997). In one view, a dissimilation force against two instances of the same structure [A] is modeled as resulting from a self-conjoined version of the markedness constraint prohibiting [A] within a particular domain, i.e., *{[A]&[A]} (e.g. Ito & Mester, 2003). While ternary conjunction is formally derivable (*{{[A]&[A]}&[A]}), Ito and Mester (2003) doubt that such constraints are ever linguistically operative.

Kawahara and Kumagai (2023) experimentally addressed this question by studying Rendaku and Lyman's Law in Japanese. Experiment 1 of that study tested whether nonce words containing two voiced obstruents are treated differently from those containing one voiced obstruent for the purposes of Lyman's Law. The results showed that no meaningful differences were observed between the two conditions, supporting the view that Lyman's Law counts only up to two i.e. Japanese phonology does not treat [D...D…D] any differently from [D...D…]. Experiment 2 of the same study re-examined the claim that two nasal consonants may reduce the applicability of Rendaku (Kim, 2020; Kumagai, 2017). The evidence for the probabilistic blocking of Rendaku by two nasals was very weak (about 3% lower than the control condition), suggesting that Japanese phonology does not avoid the [D…N…N] configuration, the result which is compatible with the thesis that phonological constraints do not count three segments.

The current study is a follow-up to Kumagai and Kawahara (2018), who investigated whether Japanese phonology disfavors sequences of nasal consonants across a morpheme boundary. In a forced-choice task, participants were asked to form a group name from two given names (an existing and well-known example being "mana-kana" from "mana" and "kana"). When one ordering created two consecutive nasals across the morpheme boundary and the other did not, participants showed some tendency to avoid the nasal sequence. However, that study had a limited number of items and did not control for the presentation order of two possible choices. Neither did it randomize the stimulus order by participant. Moreover, Kumagai and Kawahara (2018) did not directly address the question of whether phonological constraints can distinguish two nasal segments from three nasal segments (although that difference was included in the design).

The current experiment addresses two questions that remain open in light of these previous studies. The first is whether Japanese speakers avoid two consecutive nasal consonants across a morpheme boundary when forming group names, i.e., whether OCP(nasal) is operative in this domain. Given the

modest evidence from Kumagai and Kawahara (2018), we are genuinely uncertain about the answer to this question. The second question is whether the avoidance of nasal sequences is stronger when three nasals are involved (two of which reside in the same morpheme) than when only two nasals are involved. Based on the results of Kawahara and Kumagai (2023, Experiment 1), which found no evidence that Lyman's Law counts beyond two voiced obstruents, together with cross-linguistic evidence that phonological constraints do not count three segments, we expect that no such difference will be observed. Finding a difference, however, would be typologically novel and theoretically significant.

Research questions and hypotheses

Question 1. Does Japanese phonology disfavor two nasal consonants across a morpheme boundary? In other words, is OCP(nasal) operative in the domain of group-name formation? Given the modest evidence from previous studies, we are genuinely uncertain about the answer to this question.

Question 2. Does the avoidance of nasal sequences become stronger when three nasals are involved than when two nasals are involved? In other words, does Japanese phonology count beyond two for nasal dissimilation? Based on the typological generalization that segmental phonological patterns do not count beyond two (Ito & Mester, 2003; Kawahara & Kumagai, 2023, among others), we expect that no differences will be observed between the two-nasal condition and the three-nasal condition.

Methods

Overall design

The current experiment is a forced-choice task in which participants are asked to choose a more natural-sounding group name, created from two given personal names. For each trial, two group names are presented, and participants choose which ordering sounds more natural as a group name for a team consisting of two members. One ordering creates a sequence of nasal consonants across the morpheme boundary, while the other does not. The dependent variable is whether the participant chose the ordering that avoids the nasal sequence (coded as 1) or the ordering that contains it (coded as 0).

The experiment consists of two conditions crossed with two nasal types. The first condition (NN) involves pairs in which one ordering creates two consecutive nasals across the morpheme boundary (e.g., /yu**na-ni**ko/, where the boundary creates the sequence /...na-ni.../, versus /niko-yuna/, where the boundary creates /...ko-yu.../). The second condition (NNN) involves pairs in which one ordering creates three consecutive nasals across the morpheme boundary and the second name additionally contains another nasal (e.g., /yu**na-nina**/, where the boundary creates /...na-nina/ with three consecutive nasals, versus /nina-yuna/). If OCP(nasal) is operative, participants should prefer the ordering that avoids the nasal sequence in both conditions. If phonology counts beyond two (i.e. if three is worse than two), the avoidance should be stronger in the NNN condition than in the NN condition.

Each condition is tested with two nasal types (N-N and M-M), yielding four cells. Each cell contains 8 items, resulting in 32 experimental items in total. Participants also respond to three attention-check items (see Procedure).

Stimuli

The full list of experimental stimuli appears in Table 1.

Table 1. Stimuli

	conditions	item AB	item BA
2 nasals	NN	yuna-niko	niko-yuna
		rana-noka	noka-rana
		reni-naho	naho-reni
		funi-noka	noka-funi
		rine-nika	nika-rine
		yone-nachi	nachi-yone
		sino-natsu	natsu-sino
		rino-nika	nika-rino
	MM	ruma-miho	miho-ruma
		rima-mitsu	mitsu-rima
		kumi-maho	maho-kumi
		yumi-moka	moka-yumi
		rume-misa	misa-rume
		kume-moha	moha-kume
		tamo-maho	maho-tamo
		ramo-megu	megu-ramo
3 nasals	NNN	yuna-nina	nina-yuna
		kana-nena	nena-kana
		reni-nona	nona-reni
		funi-nano	nano-funi
		rune-nona	nona-rune
		yone-nano	nano-yone
		sino-nena	nena-sino
		rino-nina	nina-rino
	MMM	ruma-moma	moma-ruma
		yuma-mima	mima-yuma
		kumi-mamo	mamo-kumi
		yumi-mami	mami-yumi
		kume-mamo	mamo-kume
		rime-moma	moma-rime
		tamo-mima	mima-tamo
		rumo-mami	mami-rumo

The stimuli were those that satisfy the following characteristics. All names consist of two light CV syllables (i.e., two moras). In the NN condition, each name in the pair contains exactly one nasal consonant, and the two names are paired such that one ordering creates a sequence of two nasals across the morpheme boundary. In the NNN condition, the first name contains one nasal consonant and

the second name contains two nasal consonants, such that one ordering creates a sequence of two nasals across the morpheme boundary with a third nasal residing within the second name.

In constructing the stimuli, the following potential confounds were avoided. First, neither name in a pair should create a sequence that sounds like a productive Japanese compound (e.g., /moko-hime/, which could be interpreted as 'fluffy princess'). Second, names that are strongly associated with existing words or culturally salient terms were avoided (e.g., /nisa/, which evokes the financial term, NISA). Third, names in the NN condition were checked to ensure that they do not inadvertently contain additional nasal consonants that would make the total nasal count inconsistent with the intended condition (e.g., /mine/ contains both /m/ and /n/, resulting in three nasals rather than two). Fourth, the nasal consonants at the morpheme boundary always differ in their vowel context (e.g., /...na-ni.../, /...ma-mo.../), so that no identical CV moras are created at the boundary. This choice avoids a potential confound with identity avoidance at the CV-level, which has been shown to independently affect compound formation in Japanese (Kawahara & Sano, 2014; Kumagai & Kawahara 2018).

The two nasal types (N-N and M-M) are included in order to examine whether the pattern, if any, generalizes across different nasal consonants. Although we do not have a strong theoretical expectation that /n/ and /m/ should differ in their behavior with respect to OCP(nasal), we include the nasal type as a fixed factor in the statistical model for the sake of transparency.

Participants

We plan to collect data from 120 participants (limited to native speakers of Japanese) using the Buy Response function offered by SurveyMonkey. Data from those participants who fail the attention-check test (see Procedure) will also be excluded. We expect that we will have valid responses from about 100 participants. If the valid responses are less than 100 participants, we will collect data from additional speakers to reach the minimum of 100.

Procedure

During the instructions, participants are told that they are forming a group name for a team consisting of two members, with "mana-kana", "riku-ryuu" and “miu-mima” as actual examples. They are presented with two possible orderings in katakana. The choice of katakana is motivated by two reasons: (1) a famous group name mana-kana was written in katakana, and as such, Kumagai & Kawahara (2018) used katakana; (2) it would avoid stimuli sounding like existing compounds (although we attempted to avoid using such stimuli to the extent possible). The participants are asked to choose which ordering sounds more natural as the group name. Before the main session, they will go through one practice trial to familiarize themselves with the task.

The experiment includes three attention-check items that are structurally identical to the experimental items. As stated above, the instructions include three example pairs (mana-kana, riku-ryuu, and miu-mima) with the correct orderings indicated. These same pairs then reappear during the main session as catch trials. Participants who answer any of the three catch items inconsistently with what was shown in the instructions are excluded from the analysis.

To control for potential left-right response bias, the order of the two choices is randomized by Surveymonkey. We do *not* present two individual names in each question, whose order may bias the participants’ responses; we would instead present only two possible group names. We will explain in the

instruction, however, that the group names consist of two names, both of which are 2-letter (i.e. 2-mora) long, so that the morphological boundary in the group names should be clear. The order of the stimuli in the main session is randomized per participant by SurveyMonkey.

Statistical analyses

The results will be analyzed with a Bayesian mixed effects logistic regression model, using the brms package (Bürkner, 2017), following the analytical approach of Kawahara and Kumagai (2023).

The dependent variable is whether each item was judged to avoid the nasal sequence or not (avoidance = 1 vs. nasal sequence chosen = 0). The fixed factors are condition (NN vs. NNN) and nasal type (N-N vs. M-M), as well as their interaction. Condition is dummy-coded with NN as the reference level (NN = 0, NNN = 1). Nasal type (M vs. N) is sum-coded (−0.5, 0.5), so that the intercept represents the log-odds of avoidance averaged across the two nasal types. The model includes a random intercept for items as well as a maximal random effects structure for participants, including random slopes for condition, nasal type, and their interaction (Barr, Levy, Scheering, & Tily, 2013).

We note that with only 8 observations per cell per participant, the random slopes for nasal type and the interaction may be estimated largely from the prior rather than from the data. If the effective sample sizes (ESS) for the parameters of interest are low, we will first increase the number of iterations from 2,000 to 3,000 per chain while retaining the maximal structure, since low ESS without other diagnostic problems typically reflects insufficient sampling rather than a modeling problem. If low ESS persists and is accompanied by divergent transitions or R-hat values above 1.01, we will simplify the random effects structure by first removing the interaction slope, then the nasal type slope, retaining at minimum the random slope for condition and random intercepts.

In order words, we will start with the following model.

choice ~ 1 + condition * nasal_type + (1 + condition * nasal_type | participant) + (1 | item)

but use a simplified version if necessary

choice ~ 1 + condition * nasal_type + (1 + condition | participant) + (1 | item)

The following prior specifications will be used, following Kawahara and Kumagai (2023). A Normal(0, 1) weakly informative prior for the intercept (Lemoine, 2019) and a Cauchy prior with scale of 2.5 for all slope coefficients (Gelman, Jakulin, Pittau, & Su, 2008).

Inference criteria

With NN as the reference level in the condition factor, the intercept represents the log-odds of choosing the nasal-avoidance ordering averaged across the two nasal types. On the logit scale, an intercept of 0 corresponds to a probability of 0.5 (i.e., chance performance). Whether the intercept is credibly larger than 0 therefore tests whether OCP-nasal is operative (Question 1) in the NN condition. To assess whether OCP-nasal is operative in the NNN condition, we will test whether the sum of the intercept and the condition coefficient is credibly different from 0. The condition coefficient itself tests whether the avoidance is stronger in the NNN condition than in the NN condition (Question 2).

For each parameter of interest, we will report the posterior mean and the 95% Credible Interval. We will also report the proportion of posterior samples in the predicted direction (e.g., the proportion of posterior samples for the intercept that are positive, indicating avoidance above chance).

Following Kawahara and Kumagai (2023) who themselves followed Makowski, Ben-Shachar, Chen, and Lüdecke (2019), we will deploy an analysis using ROPE (Region of Practical Equivalence) to evaluate the evidence for null effects. Following Makowski et al. (2019), we take 0.1 of a standardized parameter, a negligible effect size according to Cohen (1988), to define the ROPE. In logistic regression models, this ROPE ranges from −0.18 to 0.18. We will calculate the proportion of posterior samples within the 95% Credible Interval that are contained in this ROPE. The ROPE analysis is particularly important for Question 2, where we are interested in demonstrating equivalence between the NN and NNN conditions, i.e., that phonology does not count beyond two.

We will interpret all results as graded evidence rather than applying a binary decision rule. This approach follows the spirit of Bayesian analysis, in which posterior distributions are interpreted as directly reflecting the likely values of the estimates (Kruschke & Liddell, 2018). We avoid the dichotomous thinking that pervades frequentist null hypothesis significance testing.

For the sake of reproducibility (Winter, 2019), the raw data, the R Markdown file, and the Bayesian posterior samples will be made available at the Open Science Framework (OSF) repository. Interested readers are welcome to examine these data, especially in ways that are not analyzed in the paper.

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